

Oracle EPM Predictive Cash Forecasting

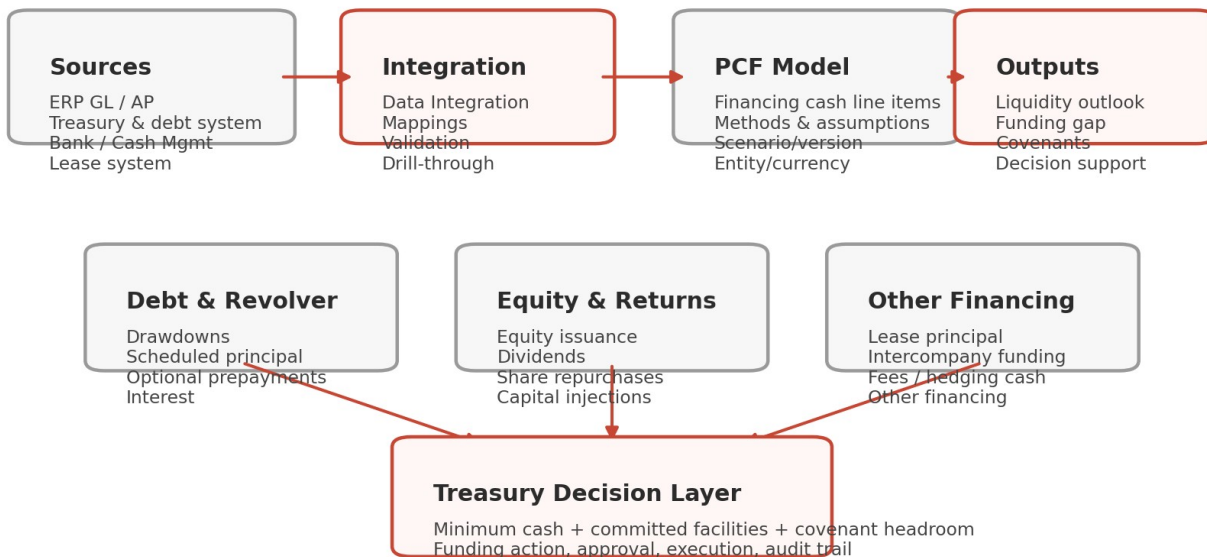
Advanced Lab Guide - Financing Cash Flow

Product Design Documentation | Implementation Guide | Training Material | Technical Reference | Troubleshooting Handbook

Audience

Advanced Oracle EPM students, treasury solution architects, finance transformation teams, implementation consultants, Cash Managers, Controllers, and application administrators.

Financing Cash Flow - Reference Architecture



BISP Trainings

1. Learning Objectives & Lab Outcomes

By the end of this lab, students should be able to design, configure, test, operate, and troubleshoot the Financing Cash Flow component of an Oracle EPM Predictive Cash Forecasting solution.

- Design financing cash line items across debt, equity, lease, intercompany and shareholder-return categories.
- Select forecast methods by line item, entity and forecast horizon.
- Model facility limits, maturity profiles, floating/fixed rates, interest timing and principal schedules.
- Build a 13-week financing cash forecast and identify funding gaps against a minimum cash buffer.
- Perform Base, Stress and Management Action scenarios and explain treasury decisions.
- Implement workflow, security, audit, job orchestration, controls and troubleshooting procedures.

Advanced-stage expectation

Students should already understand Planning dimensions, forms, business rules, Data Integration, security, scenario/version concepts, and basic treasury terminology.

Reference implementation scenario

Attribute	Lab value
Entity	US_Parent_01
Currency	USD
Forecast granularity	Weekly periodic cash forecast
Horizon	13 weeks
Minimum cash buffer	\$15.0m
Revolver limit	\$50.0m
Opening cash	\$24.0m
Debt structures	Revolver + term loan + lease liabilities
Scenarios	Base / Stress / Management Action

2. Product Design Perspective - Financing Cash Flow

Financing Cash Flow captures cash movements that change the organization's funding and capital structure. In Predictive Cash Forecasting, these line items are part of the broader Operating, Investing and Financing Cash Flow hierarchy.

Financing Cash Flow - Reference Architecture

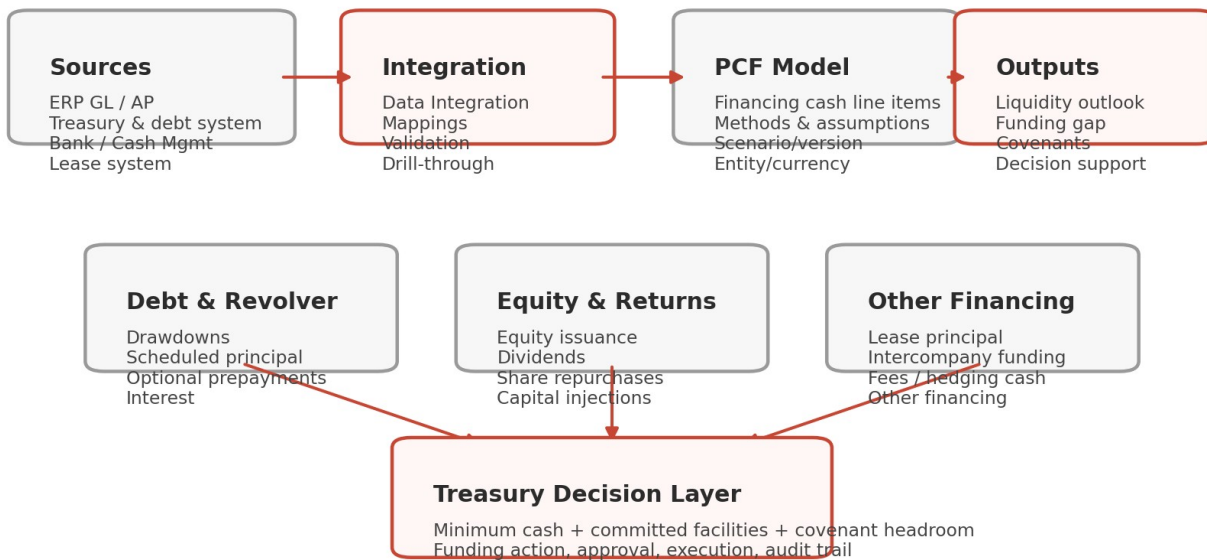


Figure 1. Financing Cash Flow reference architecture

Recommended functional line-item model

Category	Illustrative line items	Typical source / method
Debt inflows	Revolver draw; new term loan; bond proceeds	Treasury/debt system; manual or driver
Debt outflows	Scheduled principal; optional prepayment; facility fees	Debt schedule; driver
Interest	Term debt interest; revolver interest; commitment fee	Rate x balance driver
Equity inflows	Capital contribution; share issuance	Board-approved funding plan; manual
Shareholder returns	Dividend; share repurchase	Approved policy / manual
Lease financing	Lease principal payment	Lease system / ERP; driver or trend
Intercompany	Intercompany borrowing / repayment	Treasury / ERP
Other financing	Bank financing transactions not in AP/AR	Cash Management / Smart Driver

3. Oracle PCF Capability Alignment

Current Oracle documentation describes cash line items under Operating, Investing and Financing Cash Flows, and supports multiple forecast approaches selectable by line item and entity/version combination. Forecast methods can also vary by period range.

Current product behavior

Oracle PCF supports driver-based, trend-based, Smart Driver, statistical prediction and manual approaches. Smart Drivers can use transactional information from Oracle Fusion Cloud ERP; Oracle Cash Management inflows/outflows can be mapped to Financing Cash Flow in the rolling forecast grid.

Capability	Use in Financing Cash Flow
Driver-based	Calculate principal, interest, commitment fees, lease payments from assumptions.
Smart Drivers	Bring selected Cash Management financing activity into short-term forecast periods.
Trend-based	Use historical patterns for periodic recurring financing cash items where appropriate.
Prediction-based	Forecast suitable historical line items when sufficient history exists.
Manual	Board-approved dividend, strategic debt issue, one-time equity raise or override.
Period-based methods	Use near-term transactional/detail methods and later-horizon driver/trend methods.

Role-based operating model

Role	Primary responsibility
Cash Manager	Review methods, assumptions, forecast, exceptions and funding actions.
Controller	Validate consistency, reasonableness, accounting alignment and approvals.
Administrator	Configure line items, methods, integrations, rules, security and jobs.
Treasury Approver	Authorize funding actions based on policy and delegated authority.

4. Data Model & Dimensional Design

A robust design separates funding instrument, cash line item, entity, scenario, time and currency so that the forecast is explainable and auditable.

Dimension / attribute	Example members	Design guidance
Entity	Total Group > Americas > US_Parent_01	Match treasury/legal entity responsibility.
Cash Line Item	Revolver Draw, Principal, Interest, Dividend	Use meaningful cash-flow presentation members.
Scenario	Actual, Base, Stress, Action	Avoid overwriting baseline forecast.
Version	Working, Submitted, Approved	Use workflow-driven lifecycle.
Time	W1-W13 / months	Use rolling level-0 periods for forms.
Currency	USD, EUR, GBP	Define local and reporting currency logic.
Facility / instrument	REV01, TL01, LEASE01	Use custom dimension/attribute where analysis requires it.
Data source	ERP, Treasury, Bank, Manual	Preserve lineage and reconciliation.

Critical metadata properties

- Cash direction (inflow / outflow).
- Financing subcategory and reporting sign.
- Forecast method eligibility.
- Facility or instrument association.
- Entity and currency applicability.
- Approval / materiality threshold.
- Source-system mapping key and drill-through availability.

5. Source Systems, Integration & Data Lineage

Financing Data Flow & Lineage

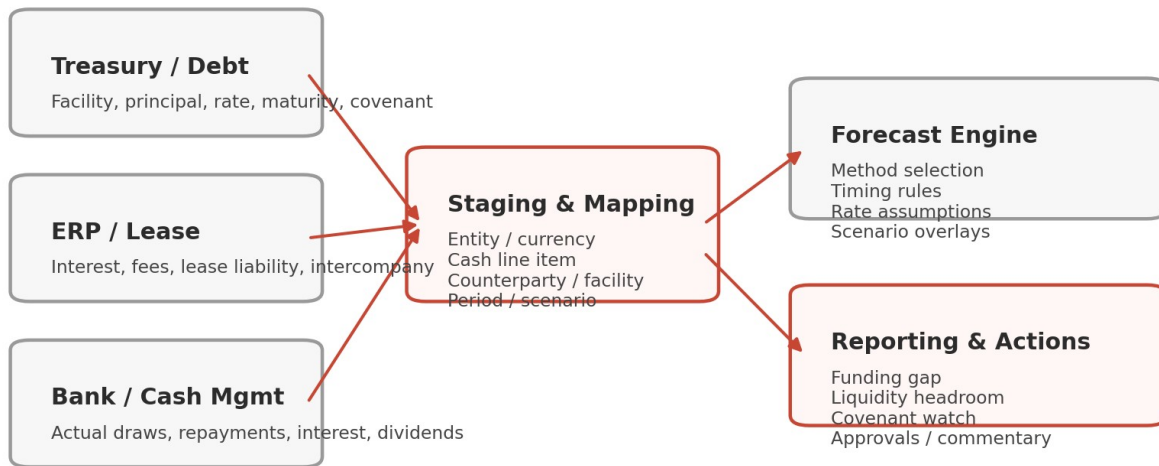


Figure 2. Financing data flow and lineage

Minimum inbound datasets

Dataset	Required fields	Control
Debt master	Facility ID, entity, currency, limit, maturity, rate type	No active facility without maturity and currency.
Debt schedule	Period/date, opening principal, repayment, draw, closing principal	Opening + draw - repayment = closing.
Rate assumptions	Index, spread, fixed rate, reset date	Freshness threshold for floating-rate curves.
Lease schedule	Lease ID, principal, interest, due date	Reconcile to lease liability / ERP.
Equity / dividend plan	Approval status, amount, payment date	Only approved actions enter Approved scenario.
Bank/Cash Mgmt	Transaction ID, value date, type, amount	No duplicate transaction IDs.

Data lineage principle

Every material financing forecast should be traceable to either a source transaction/schedule, an explicit assumption, or a documented management override.

6. Forecast Method Decision Framework

Funding Decision Tree

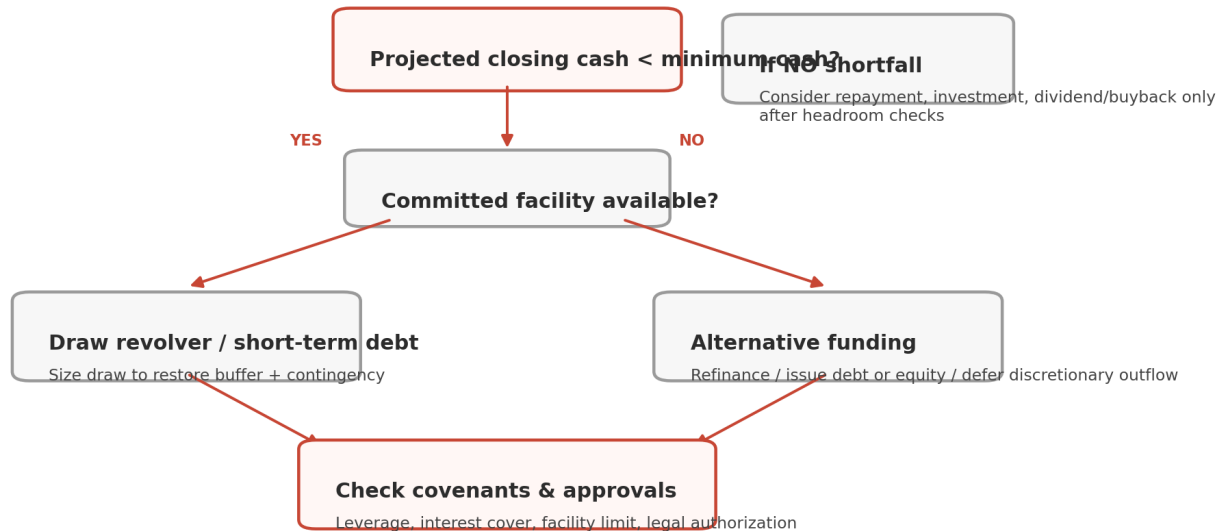


Figure 3. Funding and liquidity decision tree

Method selection by financing line item

Line item	Preferred near-term method	Later-horizon method	Why
Scheduled principal	Driver / source schedule	Driver	Contractual dates are known.
Floating interest	Driver: forecast balance x rate	Driver / scenario	Sensitive to debt balance and index rate.
Revolver draw	Rule + manual approval	Scenario/manual	Depends on projected liquidity gap.
Dividend	Manual / approved plan	Manual / policy driver	Governance-driven, not purely statistical.
Lease principal	Source/driver	Trend or driver	Often recurring, contract-based.
Other Cash Mgmt financing	Smart Driver	Trend/manual	Transactional visibility strongest near term.

7. Oracle-Inspired Configuration Mockup

Oracle-Inspired Configuration Mockup

Predictive Cash Forecasting > Periodic Cash Forecast > Financing

Entity: Version: Scenario: Horizon: Currency:

Forecast Method Assignment

Revolver Draw	Manual / Rule	W1-W4
Term Debt Principal	Driver	W1-W13
Interest Paid	Driver	W1-W13
Dividend Paid	Manual	W1-W13
Lease Principal	Trend / Driver	W1-W13
Other Financing	Smart Driver	W1-W4

[Save & Calculate](#)

Figure 4. Illustrative configuration screen - not a literal Oracle UI copy

Configuration sequence

1. Enable the periodic cash forecast granularity required by the operating model.
2. Enable the forecast methods required for financing line items.
3. Create or refine Financing Cash Flow line items and reporting hierarchy.
4. Map source accounts/transactions to financing line items.
5. Assign default methods by line item and entity/version.
6. Set assumptions for principal, rates, fees, dividend timing and liquidity thresholds.
7. Test calculations with a single entity before scaling.

8. End-to-End Financing Forecast Process

End-to-End Financing Cash Forecast Process

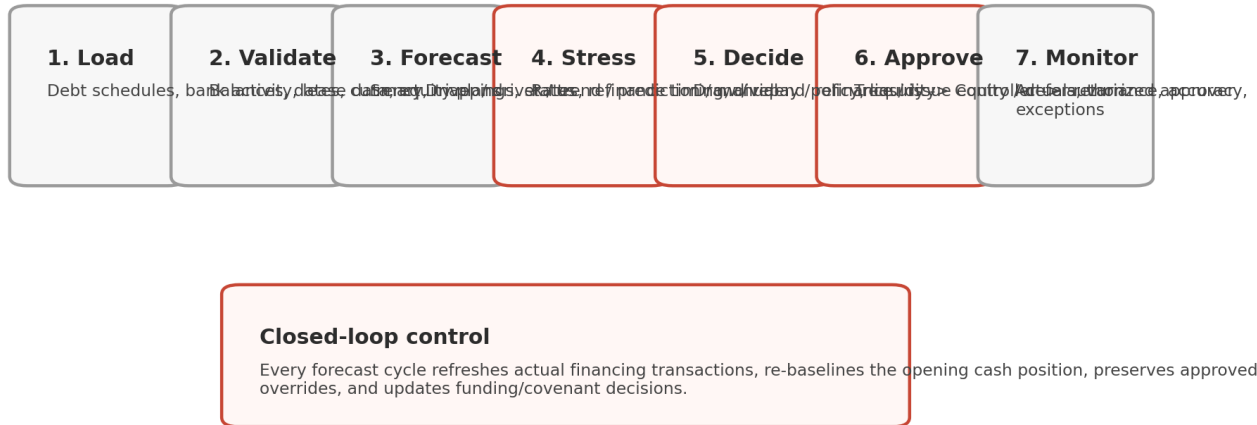


Figure 5. Financing cash forecast operating cycle

Recommended weekly cadence

Day / step	Activity	Output
T-1	Load actual bank / financing activity	Current opening cash and actual financing movements
Day 1	Load debt, lease and approved capital-action schedules	Validated source layer
Day 1	Run forecast process	Working financing cash forecast
Day 2	Review exceptions and stress scenario	Funding-gap list
Day 2	Propose funding / repayment actions	Management Action scenario
Day 3	Controller and treasury approval	Approved funding plan
Day 3+	Monitor execution vs forecast	Variance / accuracy metrics

9. Advanced Lab - Scenario Setup

The lab creates a 13-week financing forecast for US_Parent_01. Operating and Investing cash movements are assumed to be available as net pre-financing cash. Students calculate financing actions required to maintain a \$15m minimum cash buffer.

Parameter	Base assumption	Stress assumption
Opening cash	\$24.0m	\$24.0m
Minimum cash	\$15.0m	\$15.0m
Revolver limit	\$50.0m	\$50.0m
Opening revolver balance	\$8.0m	\$8.0m
Term loan principal	\$60.0m	\$60.0m
Term loan rate	6.2% annual	7.2% annual
Revolver rate	7.0% annual	8.5% annual
Dividend in W9	\$5.0m	\$5.0m unless management defers
Term principal repayment	W7: \$6.0m	W7: \$6.0m
Commitment fee	0.35% on undrawn revolver	0.35%

Lab objective

Maintain liquidity while minimizing unnecessary borrowing and preserving covenant/facility headroom. Students must explain each proposed draw, repayment or policy action.

10. Lab Step 1 - Build the 13-Week Pre-Financing Cash View

Assume pre-financing cash reflects opening cash plus forecast Operating and Investing cash movements before debt/equity actions.

Week	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
Pre-financing cash (\$m)	24.0	22.0	20.0	17.0	14.0	13.0	18.0	14.0	12.0	16.0	18.0	21.0	23.0

Interpretation

- Without financing action, W5, W6, W8 and W9 fall below the \$15m buffer.
- W9 also contains a planned \$5m dividend, increasing the policy question around shareholder returns vs liquidity.
- The funding algorithm should not automatically consume all available revolver capacity; it should size actions to the target buffer plus chosen contingency.

11. Lab Step 2 - Calculate Contractual Financing Outflows

Term-loan interest can be approximated weekly using opening principal x annual rate / 52. Revolver interest uses the applicable revolver balance. Commitment fee applies to undrawn committed capacity.

Illustrative formulas

Term interest = Principal x Annual Rate / 52

Revolver interest = Average Revolver Balance x Revolver Rate / 52

Commitment fee = Undrawn Revolver x Commitment Fee Rate / 52

Week	Term principal (\$m)	Term interest (\$m)	Principal repayment (\$m)	Dividend (\$m)
W1	60.0	0.072	0.0	0.0
W2	60.0	0.072	0.0	0.0
W3	60.0	0.072	0.0	0.0
W4	60.0	0.072	0.0	0.0
W5	60.0	0.072	0.0	0.0
W6	60.0	0.072	0.0	0.0
W7	54.0	0.064	6.0	0.0
W8	54.0	0.064	0.0	0.0
W9	54.0	0.064	0.0	5.0
W10	54.0	0.064	0.0	0.0
W11	54.0	0.064	0.0	0.0
W12	54.0	0.064	0.0	0.0
W13	54.0	0.064	0.0	0.0

For training simplicity, interest is shown weekly. Production designs should follow the actual contract payment frequency and reset conventions.

12. Lab Step 3 - Size Revolver Draws

Use a management target of \$17m during stressed weeks: \$15m minimum cash plus \$2m operating contingency. Revolver draws are proposed only when the post-contractual closing cash would fall below the target.

Week	Post-contract cash (\$m)	Proposed draw (\$m)	Revolver balance (\$m)	Action
W1	24.0	0.0	8.0	No action
W2	22.0	0.0	8.0	No action
W3	20.0	0.0	8.0	No action
W4	17.0	0.0	8.0	No action
W5	14.0	3.0	11.0	Draw
W6	13.0	4.0	15.0	Draw
W7	12.0	5.0	20.0	Draw
W8	14.0	3.0	23.0	Draw
W9	7.0	10.0	33.0	Draw
W10	16.0	0.0	33.0	No action
W11	18.0	0.0	33.0	No action
W12	21.0	0.0	33.0	No action
W13	23.0	0.0	33.0	No action

Control

Before approving a draw, validate facility availability, maturity, covenant headroom, permitted purpose and delegated authority.

13. Lab Step 4 - Base vs Stress Scenario

Stress assumptions: operating receipts are delayed, W5-W7 pre-financing cash is \$3m lower than Base, interest rates are higher, and dividend policy is reviewed.

Week	Base pre-financing cash	Stress pre-financing cash	Delta
W1	24.0	24.0	+0.0
W2	22.0	22.0	+0.0
W3	20.0	20.0	+0.0
W4	17.0	17.0	+0.0
W5	14.0	11.0	-3.0
W6	13.0	10.0	-3.0
W7	18.0	15.0	-3.0
W8	14.0	14.0	+0.0
W9	12.0	10.0	-2.0
W10	16.0	16.0	+0.0
W11	18.0	18.0	+0.0
W12	21.0	21.0	+0.0
W13	23.0	23.0	+0.0

Management Action scenario

- Defer the W9 dividend until liquidity recovers, subject to board/legal considerations.
- Draw the revolver earlier if stress signals are credible and market/liquidity risk justifies a larger contingency.
- Evaluate term-debt refinancing if the forecast shows repeated revolver dependence.
- Model rate sensitivity on floating debt and covenant metrics.

14. Covenant & Funding-Policy Controls

Financing cash forecasting becomes operationally valuable when liquidity decisions are tested against policy and covenant constraints rather than forecast cash alone.

Control	Example threshold	System behavior / workflow
Minimum cash	\$15m	Warn at \$17m; hard escalation below \$15m.
Revolver utilization	80% warning	Escalate refinancing assessment.
Facility maturity	<90 days	Flag refinancing / extension action.
Leverage covenant	<0.5x headroom	Block discretionary shareholder return without approval.
Interest coverage	<0.3x headroom	Stress rate and earnings assumptions.
Dividend policy	Board approval required	Only Approved status posts to Approved scenario.
Manual override	\$1m materiality	Require comment + supporting attachment.

15. Dashboard & Treasury Decision Cockpit

Financing Cash Flow - Treasury Dashboard Mockup

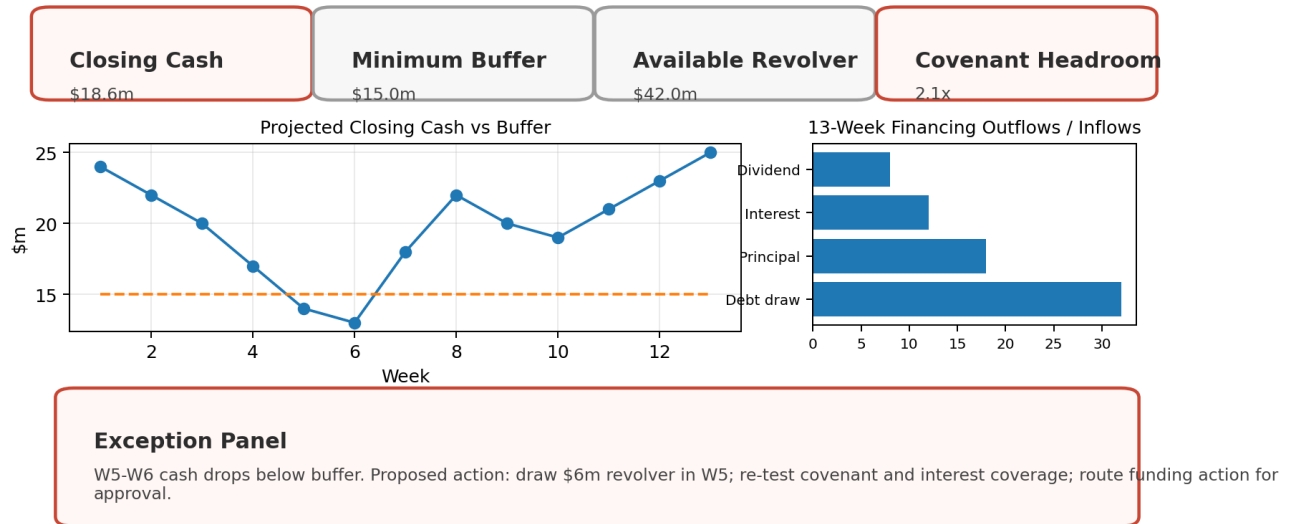


Figure 6. Financing cash flow dashboard mockup

Recommended dashboard KPIs

KPI	Definition / decision use
Projected closing cash	Detect minimum-buffer breach.
Available revolver	Remaining committed liquidity.
Net debt movement	Draws less repayments over forecast horizon.
Interest cash forecast	Rate and balance sensitivity.
Maturity concentration	Upcoming refinancing risk.
Covenant headroom	Policy / financing constraints.
Funding gap by week	Amount required to restore target cash.
Forecast accuracy / FVA	Measure whether method/override improves forecast.

16. Forecast vs Actual, Accuracy & FVA

At each roll, replace elapsed forecast periods with actual financing transactions and compare forecast to actual at the line-item and instrument level.

Metric	Formula	Example
Variance	Actual - Forecast	Actual interest 0.22 - Forecast 0.18 = +0.04m
Absolute error	Actual - Forecast	0.04m
Accuracy %	$1 - A-F / A $	$1 - 0.04/0.22 = 81.8\%$
FVA	Baseline error - adjusted error	Positive value indicates method/override improved accuracy

Root-cause dimensions

- Rate assumption error.
- Debt balance / draw timing error.
- Unplanned financing transaction.
- Incorrect source mapping or duplicate actual.
- Dividend / equity action changed after forecast cut-off.
- Calendar/date misalignment.

17. Workflow, Overrides, Security & Audit

Illustrative workflow

1. Cash Manager reviews Working forecast and exceptions.
2. Cash Manager proposes financing action in Management Action scenario.
3. Controller validates accounting, cash-flow classification and materiality.
4. Treasury Approver authorizes action within delegated authority.
5. Approved forecast is locked/published for reporting.
6. Actual execution is loaded and variances are reviewed in next cycle.

Role	Access
Administrator	Metadata, integrations, methods, rules, jobs, security
Cash Manager	Entity forecast, assumptions, proposed overrides/actions
Controller	Review, commentary, validation, submit/approve as assigned
Treasury Approver	Funding action approval; no metadata admin
Read-only stakeholder	Dashboards and approved forecast only

Audit requirement

Store who changed the amount/method, old value, new value, timestamp, comment, attachment, workflow status and source where applicable.

18. Automation, Jobs & Operational Runbook

Suggested orchestration

1. Refresh metadata / reference data if required.
2. Load actual bank and financing transactions.
3. Load debt, lease, rate and approved capital-action assumptions.
4. Run validations and reject incomplete facilities/mappings.
5. Run Periodic Process Forecast.
6. Run scenario/covenant calculations.
7. Publish dashboard and exception list.
8. Notify owners only for actionable exceptions.

Job	Frequency	Success control
Actual financing load	Daily / weekly	Source total = target total; duplicates = 0
Debt schedule load	Weekly or on change	All active facilities loaded
Rate refresh	Daily/weekly	Curve age within policy
Periodic forecast	Weekly	All target entities calculated
Exception extraction	After forecast	Owners assigned to material exceptions
Archive snapshot	Each approved cycle	Version/date retained for audit

19. Troubleshooting Handbook - Logs & Errors

Log / Error Examples

ERROR | FIN-101

Facility FAC-REV-01 maturity date missing, draw forecast blocked.

ERROR | MAP-204

Debt principal account 231500 is not mapped to Financing Cash Flow line item.

WARN | RATE-031

Floating rate assumption stale: SOFR curve older than 7 days.

WARN | COV-017

Projected leverage covenant headroom below warning threshold in W8.

INFO | JOB-200

Periodic Process Forecast completed: 14 entities, 0 rejected records.

Figure 7. Illustrative financing forecast logs

Symptom	Likely cause	Resolution
Revolver draw missing	Facility mapping or method not configured	Verify line-item method, facility status and entity mapping.
Interest forecast too high	Rate unit or frequency mismatch	Check percent vs decimal and weekly/monthly annualization.
Principal repayment in wrong week	Date/calendar mapping issue	Validate due date and forecast period mapping.
Actual financing duplicated	Multiple source loads or missing transaction key	Enforce unique source transaction ID and load rule.
Cash Mgmt activity not visible	Smart Driver/integration scope not enabled or mapped	Review integration, category mapping and forecast range.
Covenant warning false positive	Stale debt or earnings inputs	Refresh source data and recalculate dependent metrics.

20. Advanced Student Challenge

Extend the lab from a deterministic Base forecast into a policy-aware treasury model.

1. Add a second revolver with a lower rate but smaller available limit. Build a draw-priority rule.
2. Add a \$20m bond maturity in W12 and test refinance, equity raise, and partial repayment alternatives.
3. Add a floating-rate shock of +150 bps and quantify 13-week interest cash impact.
4. Create an approval threshold: any new borrowing >\$10m requires CFO approval.
5. Create a dividend deferral rule when covenant headroom <0.5x or closing cash <1.2x minimum cash.
6. Create a variance dashboard that ranks financing line items by absolute forecast error.
7. Document the data lineage for one revolver draw from source transaction through final dashboard.

Expected deliverable

Students should produce a configuration workbook, mapping specification, calculation logic, scenario results, exception report, approval workflow and troubleshooting evidence.

21. Implementation Checklist

- ☐ Confirm treasury ownership and forecast cadence.
- ☐ Define financing line-item hierarchy and reporting signs.
- ☐ Inventory facilities, debt schedules, leases, equity actions and Cash Management transactions.
- ☐ Define entity/currency/facility mappings.
- ☐ Choose method by line item and horizon.
- ☐ Configure rate, maturity, principal, fee and dividend assumptions.
- ☐ Define minimum cash and covenant warning thresholds.
- ☐ Build integrations and reconciliations.
- ☐ Test one entity / one facility end-to-end.
- ☐ Test Base, Stress and Management Action scenarios.
- ☐ Implement workflow, security and audit requirements.
- ☐ Automate jobs and exception notifications.
- ☐ Establish accuracy/FVA monitoring.
- ☐ Create support runbook and ownership matrix.
- ☐ Perform production-readiness sign-off.

22. Technical Reference & Source Notes

Key Oracle references used for product alignment

Predictive Cash Forecasting Overview

https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/cashu/cash_overview.html

About Forecast Methods

https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/caerp/cash_about_forecast_methods.html

About Smart Driver-Based Forecasting Methods

https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/caerp/cash_smart_driver_based_forecasting_methods.html

Setting up Forecast Methods and Configuring Driver Methods

<https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/pcf-forecast-methods/index.html>

Creating Predictive Cash Forecasting Applications

<https://docs.oracle.com/en/cloud/saas/planning-budgeting-cloud/pcf-create-applications/index.html>

Design note

All screens and diagrams in this guide are Oracle-inspired training mockups, not literal copies of Oracle proprietary user-interface screens. Numeric values are illustrative training data and should be replaced with organization-specific assumptions in implementation.

Summary

A mature Financing Cash Flow design links contractual financing obligations, transactional financing activity, approved capital actions and liquidity policy into one forecast-and-decision process. The strongest implementations emphasize source lineage, method appropriateness, explicit governance, scenario testing, covenant awareness and forecast accuracy - not merely cash-flow arithmetic.